There are some ancient problems on constructing geometrical objects using only a compass and a ruler (straight edge without units marking). For instance, a regular hexagon and a regular polygon of 17 sides are constructible whereas a regular heptagon and a regular polygon of 18 sides are not constructible. Using only a compass and a ruler certain geometrical constructions, particularly the following three, are not possible to construct:

- Trisecting an angle (dividing a given angle into three equal angles).
- Squaring a circle (constructing a square with area of a given circle). [Srinivasa Ramanujan has given an approximate solution in his "Note Book"]
- Doubling a cube (constructing a cube with twice the volume of a given cube).

These ancient problems are settled only after converting these geometrical problems into problems on polynomials; in fact **these constructions are impossible**. Mathematics is a very nice tool to prove impossibilities.

When solving a real life problem, mathematicians convert the problem into a mathematical problem, solve the mathematical problem using known mathematical techniques, and then convert the mathematical solution into a solution of the real life problem. Most of the real life problems, when converting into a mathematical problem, end up with a mathematical equation. While discussing the problems of deciding the dimension of a box, proving certain geometrical results and proving some constructions impossible, we end up with polynomial equations.

In this chapter we learn some theory about equations, particularly about polynomial equations, and their solutions; we study some properties of polynomial equations, formation of polynomial equations with given roots, the fundamental theorem of algebra, and to know about the number of positive and negative roots of a polynomial equation. Using these ideas we reach our goal of solving polynomial equations of certain types. We also learn to solve some non–polynomial equations using techniques developed for polynomial equations.

Learning Objectives

Upon completion of this chapter, the students will be able to

- form polynomial equations satisfying given conditions on roots.
- demonstrate the techniques to solve polynomial equations of higher degree.
- solve equations of higher degree when some roots are known to be complex or surd, irrational, and rational.
- identify and solve reciprocal equations.
- determine the number of positive and negative roots of a polynomial equation using Descartes Rule.

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